

Jay Patel . Jarvis Consulting

I am a software developer who holds a Master's degree in Big Data Analytics and possesses over 2 years of experience in web development. I've honed my skills in Python, R, SQL, and various frameworks like Angular and Spring. As a Senior Web Developer at R P Softech, I crafted a WhatsApp bot designed to facilitate bulk messaging for customer engagement purposes and managed bullion trading apps. I excel in problem-solving, algorithms, and effective teamwork. My experience at Hello Skate Mobility involved upgrading an Android app and developing an admin panel. Passionate about innovating through technology, I am seeking a role that allows me to contribute my technical expertise and creativity. My qualifications include Oracle certifications and successful project leadership in hackathons. I am eager to bring my skills to a dynamic role in web development or data engineering.

Skills

Proficient: Javascript(Angular/Node), Typescript, Firebase, Java, RDBMS/SQL, Python, Docker

Competent: GCP, Spring, PostgreSQL, Machine Learning, R, PySpark, Scala Spark

Familiar: Android, PHP, React, C/C++, Hadoop

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_JayPatel

Linux Cluster Resource Monitoring App [GitHub]: This project leverages a CentOS-based virtual machine on Google Cloud Platform (GCP) to orchestrate a VNC and Angular dashboard firewall, enabling real-time tracking of minute-by-minute memory changes. Deployed with Docker, it efficiently manages PostgreSQL, Node.js, and Angular, while Bash scripts handle table initialization and data parsing from tools like vmsat and lscpu. Crontab ensures regular data insertion. With a focus on RDBMS (PostgreSQL), CentOS, Bash, and Docker, this initiative integrates diverse technologies for seamless data tracking and visualization, enhancing system management through robust analytics.

SQL Practice [GitHub]: This repository offers a platform for honing SQL skills through practical exercises using a sample database schema. It provides a comprehensive set of SQL queries covering data manipulation, retrieval, aggregation, and joining. Aimed at individuals seeking to improve their SQL proficiency, including students, database administrators, and software developers, the project employs PostgreSQL as the relational database management system and Docker for easy setup and deployment. With a focus on interactive learning, users can practice real-world scenarios, enhancing their understanding of SQL concepts and syntax through hands-on experience and structured exercises.

GREP App [GitHub]: The JavaGrep application is a command-line tool for searching text files using regex patterns. It's built with core Java functionality and libraries, offering users the ability to specify patterns, directories to search, and output files. Features include regex pattern matching, file I/O, and logging. The app aims to enhance SQL skills through practical exercises, suitable for students, DBAs, and developers. Potential improvements include multithreading for performance, advanced search options, and enhanced error handling. The app can be dockerized for easy distribution, and manual testing ensures accuracy.

Stock Quote Application [GitHub]: The Java Stock Quote Application is a Java 11 tool that allows users to manage stock information from a PostgreSQL database. Developed with Maven and Docker, it fetches data from the Alpha Vantage API, parsing it into JSON format and storing it in the database using QuoteDAO. The application follows a service layer architecture, including Entity, DAO, and service layers, with a controller layer for user interaction. Unit testing covers DAO methods, while integration testing uses Mockito for database and API calls. Manual testing involves inspecting database changes against expected values. This application provides a robust solution for managing and interacting with stock data efficiently.

Highlighted Projects

Facial Expression Detection [GitHub]: Explore the depth of human emotions through our Facial Expression Recognition project. Implemented in Python, it employs advanced libraries like Keras, sci-kit-learn, and OpenCV. I crafted a convolutional neural network featuring pooling layers and ReLU activation functions, unveiling a sophisticated approach to decoding facial expressions.

Market Basket Analysis - Instacart [GitHub]: Analyzed over 3 million transactions from the Instacart dataset, employing market basket analysis for insights into customer buying habits. Utilized association rule mining algorithms, including Apriori and FP-growth, to reveal significant trends and identify commonly purchased product combinations.

My Portfolio [GitHub]: I created my portfolio website, showcasing proficiency in Full Stack Development, AI, and more. Crafted using HTML/CSS and powered by JavaScript, the site navigates seamlessly, reflecting my commitment to excellence. Dive into my journey, skills, and diverse projects. Explore testimonials and connect for collaborative opportunities. Elevate your browsing experience with a clean, responsive design. Browse my tech journey, witness my skills, and let's connect!

Professional Experiences

Software Developer, Jarvis (Jan 2024-present): As a Software Developer at Jarvis Consulting Group, spearheaded the development of the 'Linux Cluster Resource Monitoring App.' Orchestrated a CentOS-based virtual machine on Google Cloud Platform, implementing a VNC and Angular dashboard firewall for real-time memory tracking. Employed Docker to streamline PostgreSQL, Node.js, and Angular deployment, optimizing system performance. Executed Bash scripts for table initialization and data parsing, ensuring efficient utilization of RDBMS (PostgreSQL). Implemented crontab for scheduled data insertion, harmonizing diverse technologies for seamless data tracking and visualization. Jarvis's role significantly contributed to enhancing system management through robust analytics.

Senior Web Developer, R P softech (Sep 2020 - December 2021): I spearheaded the development of a WhatsApp Bot, leveraging Electron and Angular to create a seamless, user-friendly interface. This interface significantly improved customer engagement by 30% through efficient bulk messaging capabilities. The management of the database was executed seamlessly with Node.js. Additionally, I engineered two bullion trading apps, implementing a hybrid database with both Sequel and MongoDB. The integration of a web socket between Node and Angular facilitated rapid rate updates. Furthermore, I was involved in the end-to-end development of Accounting software, incorporating Angular for the frontend, Node.js for API interaction, and ensuring a keyboard-driven, auto-backed-up system for day-to-day transaction tracking.

Web Developer & Head of Interns, Hello Skate Mobility (July 2019 - Sep 2020): Engineered an advanced admin panel employing Spring and Firebase, empowering administrators with real-time oversight of scooter positions and statuses. Implemented cutting-edge features, including geolocation tracking and instant alerts for status changes. Leveraged Firebase's robust database and messaging services to craft a scalable and efficient solution. Additionally, elevated an existing native Android app to a hybrid version using Angular and Firebase, optimizing performance by an impressive 30% through Cordova for native functionality integration.

Education

Trent University (Jan 2022- May 2023), Master's of Big Data Analytics, Applied Modelling and Quantitative Methods - GPA: 3.8/4.0

Gujarat Technological University (2017-2021), Bachelor of Engineering, Information Technology - Facilitated an Android workshop - Represented the college in a national-level coding competition.

Miscellaneous

- Oracle Certified Expert, Java EE 6 Web Component Developer
- Oracle Certified Professional, Java SE 6 Programmer
- Databricks Certified Data Engineer Associate
- Achieved 1st Place in the Hackathon hosted by Jarvis Consulting Group. And Teranet Inc.
- Competitive Coding
- Tactical first-person shooter games